



Post-Quantum Cryptography Migration

Part 2 - Silicon & Firmware: Firmware & Root of Trust

Compiled by the Spaced Research Ledger

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Every computer decides what software to trust before the operating system even starts. That decision rests on cryptographic signatures burned into firmware, boot chains, and security chips. These are the hardest components to fix after they ship, so they carry the earliest post-quantum deadlines in the whole migration.

This report covers them in four sections. Section 1 covers the boot chain and Section 2 the TPM security chip and its standards. Section 3 covers attestation protocols and silicon roots of trust, and Section 4 the signing of firmware updates.

Two things changed decisively this year. The Trusted Computing Group published the first TPM specification with post-quantum algorithms built in, and followed it with buyer guidance for telling a quantum-ready TPM from a marketing claim. And the standards work for post-quantum boot chains moved from discussion to concrete proposals, with the UEFI Forum publishing draft specification changes and reference firmware code gaining real ML-DSA support.

One widespread confusion deserves clearing up front. Microsoft rotated its Secure Boot signing certificates in June 2026 because the old ones expired, and some coverage presents that as quantum readiness. Microsoft's own engineers say it is not. The actual post-quantum transition for Secure Boot is planned for around 2030, and Section 1 covers the disagreement.

Meanwhile the update-signing world is not waiting for the boot chain. Android now verifies its boot sequence with a post-quantum algorithm, and Microsoft plans post-quantum code signing by default in 2027. At least one storage vendor already ships hard drives with dual classical-plus-quantum firmware signatures.

1. UEFI & Boot Chain

When a computer powers on, its firmware checks a chain of cryptographic signatures before handing control to the operating system. That process, called Secure Boot on most machines, is governed by the UEFI specification and a small set of signing certificates, most of them controlled by Microsoft. If those signatures can be forged by a quantum computer, everything the machine runs afterward is built on sand. This section covers how the boot chain's specifications, reference code, and certificates are preparing.



Recent Developments

The specification work turned concrete this year, even though no product ships post-quantum Secure Boot yet. The UEFI Forum's Security Sub-Team published a whitepaper proposing changes for a future "UEFI.next" specification [1]. Intel and Microsoft editors wrote it, with contributors from a dozen major firmware and hardware vendors [1].

The reference firmware moved too. TianoCore EDK II, the open-source base most UEFI firmware builds on, merged ML-DSA signing and verification APIs into its cryptography package in July, with unit tests [2]. A follow-up commit hardened parameter validation in the verification path a week later [3].

The signing chain above the firmware is moving on the same algorithm. Linux kernel patches under review add ML-DSA support to kernel module signing, extending the post-quantum chain from firmware into operating-system code signing [4]. On Windows, Microsoft shipped hybrid ML-KEM key exchange for TLS in a July cumulative update as part of its Quantum Safe Program. That protects connections, not the Secure Boot trust chain itself [5].

Current State

The boot chain's biggest event in 2026 has nothing to do with quantum resistance, and keeping the two apart takes some care. Microsoft's original 2011 Secure Boot certificates expired on June 27, 2026, and are being replaced by 2023-generation certificates. Existing signed binaries keep booting, but new binaries can no longer be signed with the old key [6]. Linux distributions handled it by shipping boot loaders signed by both generations, with some platforms moving to the 2023-only signer [7][8]. In the same period Microsoft revoked roughly a dozen old signed boot loaders that could bypass Secure Boot, ordinary operational churn in the revocation infrastructure [9].

The genuine post-quantum plan sits years out. Microsoft engineers said at a March 2026 session that the post-quantum transition for Secure Boot certificates on future hardware is planned for around 2030 [10]. What exists today is preparation. Microsoft enabled ML-DSA certificate issuance in Active Directory Certificate Services on Windows Server 2025, explicitly as a way to evaluate post-quantum signing workflows [11]. It requires a new parallel certificate hierarchy and is separate from signing the boot chain [11].

The UEFI whitepaper's proposals show what the eventual transition requires. They include allowing multiple signatures per boot image, so a revoked algorithm does not invalidate an otherwise-valid one [1]. They add storage of certificate hashes instead of full certificates, because post-quantum certificates are far larger [1]. And they define a new firmware table letting the operating system query which algorithms the firmware actually supports [1].

The design deliberately defines no new cryptography, reusing the IETF's certificate and signature formats instead [1]. The products must not ship any of it until the specification is formally adopted [1].

Two dependencies sit outside UEFI's control. The composite classical-plus-quantum signature format the whitepaper references is still an IETF draft, so implementations depending on it wait on standardization [12]. And government guidance, chiefly NSA's CNSA 2.0 and NIST's migration report, sets the backdrop the whole effort designs against [13]. The UEFI security team's public tracking issue has been open since 2022 [13].

The open-source alternatives lag the specification work. coreboot's verified boot still documents an RSA-only root of trust with no post-quantum alternative [14], and its two most recent releases mention no post-quantum work at all [15]. The shim boot loader's repository shows no ML-DSA or SLH-DSA changes either, with current effort going to revocation handling and other hardening [16].

Unresolved Conflicts

The one live disagreement is about framing, and it matters because it shapes buyer expectations. Some vendor and news coverage presents Microsoft's 2026 certificate rotation and post-quantum readiness side by side as one trust-modernization story. Microsoft's own engineers and Red Hat's technical guidance treat the rotation as unrelated to quantum resistance, with the real Secure Boot transition around 2030 [17]. Microsoft's published transition plan for its signing infrastructure would settle which reading buyers should act on [17].

2. TPM & TCG Specifications

The Trusted Platform Module is the small security chip in most business computers. It stores keys, measures what software booted, and proves a machine's identity to networks and services. The Trusted Computing Group writes its specifications. Because TPMs are usually soldered to the motherboard, whatever algorithms they ship with are what the machine keeps, which makes the specification's algorithm choices unusually consequential.

Recent Developments

The specification milestone arrived in March. The Trusted Computing Group released TPM 2.0 Library specification version 185, its first with post-quantum capabilities, adding ML-KEM and ML-DSA [18]. The final document published on March 12 after two rounds of release-candidate review [19]. TCG called it a critical milestone for the organization and for vendors building on its standards [20].

Guidance for buyers followed in August. New TCG requirements define what counts as a "PQC-ready TPM," anchored to the PC Client Platform TPM Profile 1.07 built on the version 185 library [21]. A separate "PQC-upgradable" designation covers modules that can get there by update [21]. The point, as coverage noted, is giving buyers a way to test vendor claims instead of taking them on faith [22].

Hardware is starting to follow the ink. SEALSQ is sampling its TPM185 post-quantum module [23]. Its revised roadmap targets FIPS 140-3 submission around February 2027, TCG certification around April 2027, and first revenues by the end of 2026 [24]. The DICE device-identity specifications entered a fresh public-review round in July [25].

Current State

Version 185 makes specific, conservative algorithm assignments. ML-KEM applies to the TPM's Endorsement Key for long-term confidentiality, and ML-DSA to Attestation Keys [18]. The release also adds Curve25519 and Curve448 for compatibility with protocols that already use them [26]. Software support exists ahead of silicon: wolfSSL's wolfTPM library implements the new command set, including the encapsulation and signing

commands [27]. The same vendor announced a firmware TPM implementing the full version 185 command set, across every ML-DSA and ML-KEM parameter set, for embedded platforms [28].

The surrounding TCG portfolio is adjusting at its own pace. The DICE attestation architecture, whose key-derivation model is independent of which signature algorithm signs its certificates [29], stands at version 1.2 [30] with active errata maintenance [31]. Its interoperability binding to the SPDm protocol does not itself add post-quantum algorithms [32]. The Platform Certificate Profile now states the root of trust need not be the TPM at all, admitting DICE or MARS instead [33].

3. Attestation Protocols & Silicon RoT

Attestation is how one piece of hardware proves to another what it is and what it is running. Data centers depend on it: a server checks its network card, a cloud checks its servers. The proofs are signatures from a root of trust, a minimal piece of silicon whose only job is to be trustworthy. This section covers the main open efforts, the SPDm protocol and the Caliptra and OpenTitan roots of trust, and the first security research against post-quantum hardware.

Recent Developments

The SPDm protocol is taking public input on its post-quantum future. The DMTF working group published its plan for hybrid classical-plus-quantum support in the forthcoming SPDm 1.5 [34] and opened formal feedback with an August 31, 2026 deadline [35]. Meanwhile SPDm 1.4.1 published in July, and the libspdm reference implementation reached version 3.8.2 [36]. A community Rust implementation reported full SPDm 1.4 support including ML-DSA and ML-KEM, cross-tested against libspdm, plus a new use securing key exchange for live migration of confidential-computing workloads [37].

On the silicon side, SEALSQ's QVault TPM roadmap revision also lands here: engineering samples available now, certifications targeted through 2027, first revenues expected by the end of 2026 [38]. TCG's PQC-ready TPM guidance applies to the measured-boot chain these devices anchor [21].

OpenTitan's project news came in a cluster in late August. The published roadmap introduces a second-generation top-level design, Earl Grey 2, with hardened ML-KEM and ML-DSA support, CHERI memory safety, and RRAM, targeting initial implementation during 2026 [83]. A companion top-level called Peppermint, an integrated root of trust aimed at area-constrained chips such as wearables, was announced the same week [84]. Caliptra's own cadence continued: the first Caliptra 2.1 Subsystem silicon powered on in mid-August [85], after FIPS certification of the 2.x cryptographic algorithms completed in July [85].

Current State

Caliptra, the open-source silicon root of trust, is the furthest-along post-quantum hardware. It authenticates firmware with a dual-signature scheme pairing ECDSA P-384 with ML-DSA-87 through its layered DICE identity architecture [39]. Since version 2.0 the specification allows those dual signatures to satisfy FIPS 204 and CNSA 2.0's highest category, plus ML-KEM alongside classical key exchange [40]. Version 2.1 integrates Adams Bridge 2.0, an accelerator providing ML-DSA and ML-KEM with side-channel countermeasures [41].

Adams Bridge itself is described as the first open-source lattice-cryptography hardware accelerator, unifying both algorithms in one pipelined datapath with measured efficiency gains at 5nm [42].

The project was founded in 2022 by Microsoft, AMD, Google, and Nvidia [41].

OpenTitan is the root of trust that already ships post-quantum boot. Its first production chip, Earlgrey, verifies firmware with hardware-accelerated SPHINCS+ signatures, compliant with the FIPS 205 SLH-DSA standard [80]. Nuvoton began mass production in February 2025, with the chips headed for Chromebooks and Google’s own data-center servers [81]. By its partners’ account this is the first production security chip whose secure boot rests on a NIST post-quantum signature [80].

The signing side is next. OpenTitan partners approved hardware extensions to the project’s big-number accelerator for ML-DSA and ML-KEM [80]. The additions are eight times the data memory, a direct path to the hardened hashing block, vectorized polynomial arithmetic, and automated mask conversion [80]. The mask hardware matters because side-channel protection otherwise dominates cost, with masking overhead near 90% without it [80]. The changes are planned to complete in 2026, with work under way in the public repository since February [80].

Commercially, ZeroRISC and research partners including the Max Planck Institute for Security and Privacy released an open-source cryptographic stack for this silicon [82]. It ships hardware-accelerated ML-KEM, ML-DSA, and SLH-DSA, with reported speedups of six to nine times [82].

SPDM’s post-quantum arc has broad institutional weight behind it. Six standards bodies jointly announced the protocol’s evolution toward post-quantum support aligned with CNSA 2.0 [43]. The libspdm reference already supports ML-DSA, SLH-DSA, and ML-KEM alongside its classical set [44]. The protocol’s key-exchange mechanism entered FIPS 140-3 implementation guidance in April 2026, giving validated modules a recognized path to using it [45]. The first industry feedback round on post-quantum SPDM dates back to mid-2024 [46].

Commercial measured-boot products are marketing post-quantum support today. Dyber’s QuantaTPM advertises TCG certification, ML-KEM key establishment, and CNSA 2.0 alignment, with FIPS validation described as in process [47]. SEALSQ’s QVault targets smart meters and grid controllers with ML-KEM and ML-DSA alongside classical algorithms [48]. A published research taxonomy frames the whole field’s central challenges as larger keys, larger signatures, and higher compute cost, with open questions in migration strategy and formal modeling [49].

Unresolved Conflicts

The first security research against post-quantum root-of-trust hardware has produced a genuine open question. Researchers published a power-analysis side-channel attack against the ML-DSA implementation inside Caliptra’s Adams Bridge accelerator, the first documented attack of its kind [50]. A later analysis re-examined the accelerator’s partial masking countermeasures and questioned whether its claim that unprotected layers are impractical to exploit holds up [51]. Sources disagree on how resolved the concern is. An independent evaluation of the countermeasures, or a revised design, would settle it [51].

4. OTA Update Signing

Every device that updates itself over the air checks a signature before installing new firmware. That signature is the last line of defense against malicious updates, and its algorithm is baked into devices that may sit in the field for decades. This is why firmware signing carries the earliest post-quantum deadline of any category in United States government guidance. This section covers the signing schemes, the tools, and who has actually switched.

Recent Developments

The standards and platform announcements now come monthly. An IETF draft on constrained devices recommends the stateful hash-based schemes LMS and XMSS for firmware signing [52]. It notes the relevant working group may add ML-DSA and SLH-DSA to update formats later [52]. A companion draft defines composite classical-plus-quantum signatures for the JOSE and COSE formats used in artifact and firmware signing [53]. Microsoft plans to move Windows production signing to post-quantum by default in 2027, after replacing its certificate authorities in 2026 [54].

Toolchains gained concrete support through August. Renesas's port of the MCUboot boot loader now verifies all three ML-DSA parameter sets [55]. STMicroelectronics's quantum-resistant TPM uses LMS for secure firmware updates as a hardware root of trust [56]. The SWUpdate embedded updater added hybrid classical-plus-quantum signing with multiple signers, including ML-DSA certificates [57].

Current State

Government policy makes this the most urgent corner of the migration. NSA's CNSA 2.0 advisory set software and firmware signing as its earliest transition category: begin immediately, prefer the new algorithms by 2025, use them exclusively by 2030 [58]. Commentary attributes the tight timeline to the fact that firmware roots of trust are the hardest components to update after deployment [59]. NSA has urged LMS and XMSS adoption since 2022, and validated hardware has been available for over a year [59].

The practical pattern is dual signing. Vendors sign new firmware with both a classical and a post-quantum algorithm in parallel [60]. They publish the post-quantum key even before field devices can verify it, so integrity can later be proven against a quantum-capable adversary [60].

Industry guidance treats LMS and XMSS as the deployable choice today [61]. ML-DSA is the likely long-term preference once tool validation catches up, because it needs no one-time-key state tracking [61]. One embedded-security assessment places lattice signatures only in pilot use in secure boot chains, with hybrid signing expected to dominate through 2030 [62].

Real deployments exist now. Android 17 integrates ML-DSA into Android Verified Boot and extends it to app signing through the Keystore and Google Play [63]. Western Digital adopted ML-DSA-87 dual-signed with RSA for secure boot and firmware signing in enterprise hard drives, with its post-quantum signing infrastructure deployed in 2025 [64]. Infineon's Optiga TPM firmware updates are already signed with XMSS, which the chip verifies directly with sub-second verification [65]. A defense contractor demonstrated post-quantum firmware-update signing on a memory-safe chip [66], and a public demonstration integrated ML-DSA and SLH-DSA into ESP32 secure boot and update flows [67].

The boot-loader ecosystem is split. wolfSSL's wolfBoot supports LMS [68], XMSS [69], and ML-DSA [70]. The upstream MCUboot documentation still describes only classical algorithms [71], even as the Renesas port adds ML-DSA, so support depends on which build a vendor ships [55]. The UEFI whitepaper's multiple-signature and certificate-hash proposals extend the same modernization to PC-class update signing [72].

Signing services and hardware complete the chain. AWS added ML-DSA signing to its key management service, aimed explicitly at firmware and code signing where signatures cannot be changed after deployment [73]. It then extended the capability to full certificate hierarchies through its private CA [74].

PQShield sells secure-boot silicon supporting LMS for fast boot verification, ML-DSA for general signing, and ML-KEM for encrypted-image keys [75]. In vehicles, researchers have benchmarked post-quantum signatures against the Uptane update framework the automotive industry standardized on [76]. A dedicated survey covers vehicle migration [77], and commercial workshops sequence the transition by safety criticality under the industry's regulatory obligations [78]. Trade coverage sums up the state plainly: post-quantum firmware signing is no longer theory, with standards bodies and silicon vendors pushing it into embedded toolchains [79].

About This Report

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